Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
MAKEUP EXAMINATIONS, DECEMBER 2024

REAL TIME SYSTEMS [ELE 4064]

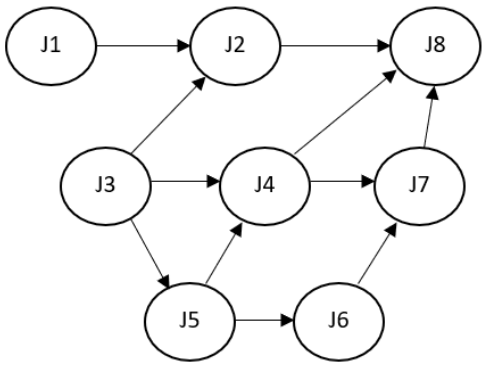
REVISED CREDIT SYSTEM

Time: 3 Hours **Date: DECEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Develop a schedule for the job set consisting of 8 jobs (J_1-J_8) on a uniprocessor timeline and analyze its preemptive performance. Consider the following release time (ms) and execution time (ms) of each job: J_1(r1=1;e1=2), J_2(r2=0,e2=3), J_3(r3=1,e3=2), J_4(r4=2,e4=1), J_5(r5=3,e5=4), J_6(r6=2,e6=3), J_7(r7=7,e7=2), J_8(r8=4,e8=2). The precedence graph is given below. Consider phase ($\emptyset = 9$) ms for Job 6. Assume higher priority for jobs with higher index. ***(02)***



1B. Analyse and schedule a real-time system with a pre-emptive periodic task set given in the table using DMA on a timeline of [0-250]ms. Verify whether all the tasks meet their deadlines. Assume deadline is relative to the phase. ***(04)***

Tasks	Phase (ms)	Period (ms)	Execution (ms)	Deadline (ms)
T1	50	50	25	100
T2	0	62.5	10	20
T3	0	125	25	50

1C. Analyze using relevant tests whether the following set of periodic real time tasks is schedulable under RMA on a uniprocessor. Validate your answer by scheduling the task set on the timeline of [0-109]ms. ***(04)***

Tasks	Period (ms)	Execution Time (ms)	Deadline (ms)
T1	30	14	30
T2	40	11	40
T3	120	18	120

2A. A real time system with the following task set must be pre-emptively scheduled on a uniprocessor system. ***(03)***

Tasks	Release Time (ms)	Period (ms)	Execution Time (ms)	Deadline (ms)
T1	0	2	0.4	2
T2	1	4	1	4
T3	0	5	1.5	5

i) Analyze whether the task set can be feasibly scheduled using Earliest Deadline First (EDF) with appropriate justification

ii) If the parameters of the system are changed, such that T3 has release time of 0ms, period of 8ms, execution time of 4ms and deadline of 8ms. Verify whether the new system task set can be scheduled using EDF with appropriate justification.

2B. A real time system has three independent periodic tasks defined w.r.t their period (ms), execution time (ms) and deadline (ms) as: T1(4,1), T2(5,2), T3(20,5). Compute a suitable frame size. Show all intermediate computations. ***(04)***

2C. A real time system consists of the following periodic and aperiodic tasks. Periodic tasks are defined w.r.t their period (ms) and execution time (ms) as: T1(2.5,1), T2(9,3); Aperiodic task: A1 arrives at t = 0.1 ms with execution time of 0.9ms; A2 arrives at t = 2.7ms with execution time of 1ms. ***(03)***

Analyze and schedule the following tasks using RMA and deferrable server with period (ms) and server budget (ms) as (2.4,0.6). Determine the response times of A1 and A2. Also check whether the periodic tasks meet their deadline.

3A. Analyze and schedule the task set shown in the below table using Least Slack Time (LST) scheduling method. Show the time-line [0 to 16ms] of the schedule. Show all the intermediate steps in your calculations for schedule. (Assume secondary priority rule of higher index as higher priority) ***(04)***

Task	Release time (ms)	Execution time (ms)	Deadline (ms)
T1	0	5	10
T2	2	8	16
T3	4	3	12

3B. Consider a real time system with the task set below consisting of three periodic tasks T1, T2, and a aperiodic task A1. ***(04)***

Tasks	Release Time (ms)	Period (ms)	Execution Time (ms)
T1	0	3	1
T2	0	8	4
A1	0.3	-	1.8

Analyze and schedule the task set on the timeline until the aperiodic task completes its execution using Rate Monotonic (RM) Scheduler considering the below approaches.

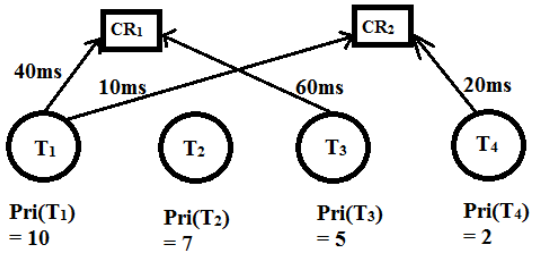
i. Without slack stealing

ii. With slack stealing

Evaluate the response times of aperiodic tasks in both the cases and the completion time of the first job of task T2.

3C. Outline the working principle of Highest Locker protocol (HLP). Illustrate with an example how HLP overcomes the drawbacks of Priority Inheritance Protocol (PIP). ***(02)***

4A. A system has four tasks: T1, T2, T3 and T4. Two critical resources are shared: CR1 and CR2. The priority values of each task are shown in the below figure. Larger priority values indicate higher priority. The resource requirements of these tasks and the duration for which the tasks need the three resources are also shown in the figure. Analyse the different types of inversions that each task might have to undergo for the given task set. ***(04)***



4B. The specifications of a set of ten periodic real-time tasks (T1-T10) are shown in the below table. Assume that the task set need to run on a multi-processor system with four processors. Analyze and allocate the tasks to the processors using next-fit algorithm and utilization balancing algorithms. ***(04)***

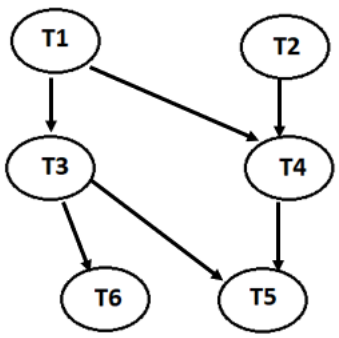
Task	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
Execution(ms)	5	5	3	1	8	11	1	3	9	17
Period(ms)	10	22	22	24	30	40	50	55	70	75

4C. Considering a real time system with appropriate task set and system characteristics, analyze with proper justification the following situations with respect to priority ceiling protocol (PCP) ***(02)***

i) Avoidance of chain blocking
ii) Avoidance of deadlocks

5A. Analyze and schedule the below task set using Deadline Monotonic Algorithm (DMA) scheduler on timeline [0-100]ms considering the precedence constraints shown in the below figure. [Assume period is equal to deadline]. ***(04)***

Tasks	Release Time (ms)	Execution Time (ms)	Deadline (ms)
T1	0	1	10
T2	6	2	15
T3	0	3	25
T4	0	1	40
T5	0	5	65
T6	5	5	70



5B. With appropriate justification and example summarize the requirements for an operating system to be called as a real time operating system. ***(02)***

5C. A network designed using IEEE802.4 protocol has three nodes: N1, N2 and N3. ***(04)***

N1 needs to transmit 1 MB of data every 300 msec.

N2 needs to transmit 1.2 MB of data every 500 msec.

N3 needs to transmit 2 MB of data every 200 msec.

Select a suitable Target token rotation time (TTRT) for the network and determine the token holding time for each node.